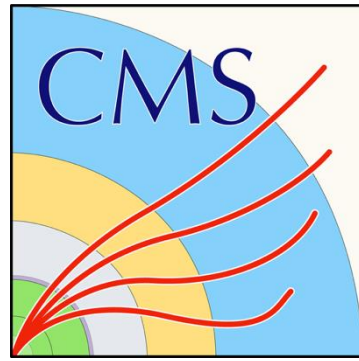




PRINCETON
UNIVERSITY

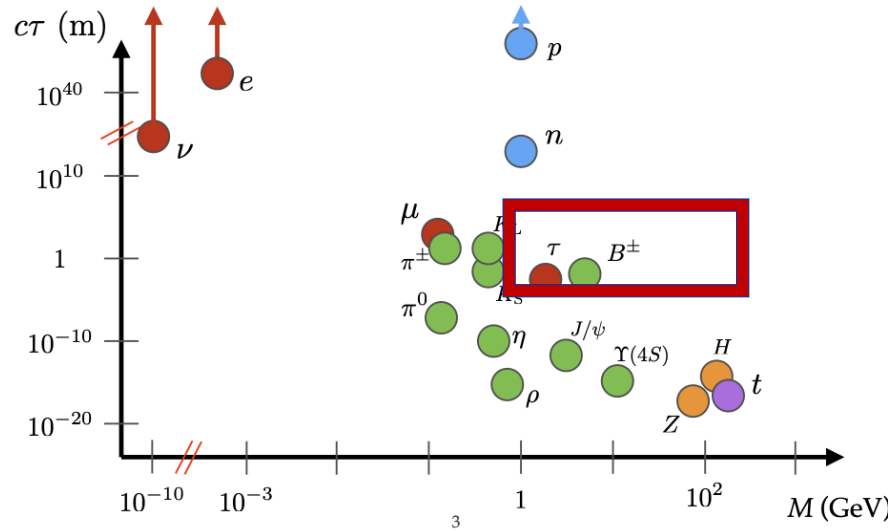
Caltech



Search for Long-Lived Particles with HCAL Segmentation in CMS at the Large Hadron Collider

Katia Avanesov, Harvey Newman, Kiley Kennedy, Gillian Kopp, Cristian Peña, Chris Tully, Si Xie

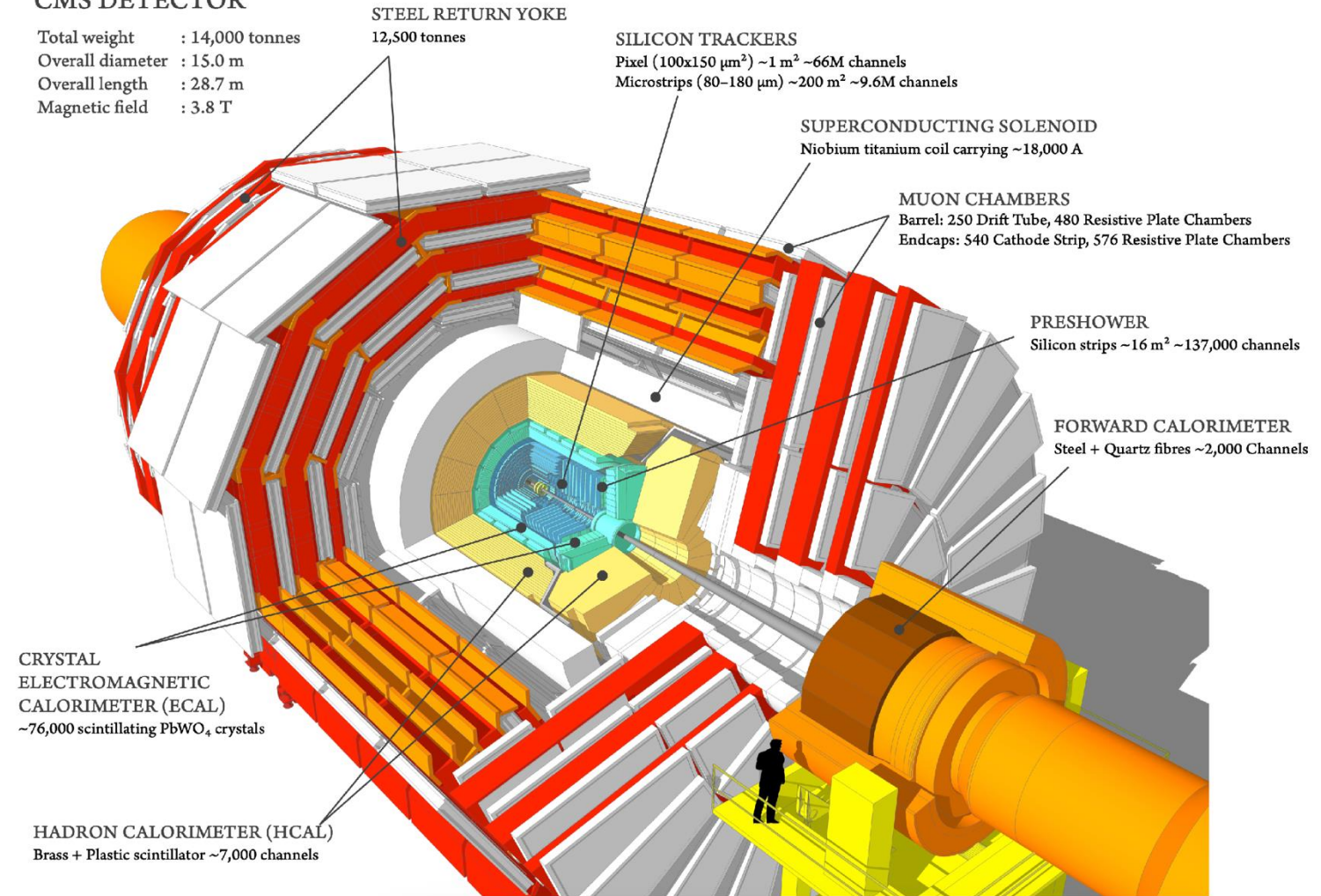
Long Lived Particles (LLPs)



- LLPs have lifetimes greater than 0.1 nanoseconds
- LLPs feature in both the Standard Model and predicted in numerous Beyond the Standard Model (BSM) theories

CMS DETECTOR

Total weight : 14,000 tonnes
Overall diameter : 15.0 m
Overall length : 28.7 m
Magnetic field : 3.8 T



*Evenly distributed
calorimeter hits*

Many tracks

Prompt jet

2.95 m

HICAL

1.77 m

ECAL

1.29 m

Tracker

-
- The diagram illustrates the ATLAS detector's calorimeter system, specifically the High Calorimetry (HCAL) Barrel and Endcap. The HCAL Barrel is shown as a large, rectangular structure with a grid of cells, colored in pink, green, and yellow. The HCAL Endcap is a smaller, curved structure at the top, also with a grid of cells, colored in pink, green, and yellow. A dashed line indicates the path of a particle through the detector, with labels for the barrel length l_b and the LLP path length l_{LLP} . The expected time delay $t_{expected} = \frac{l}{c}$ is also shown.

$$\Delta t = \frac{l_{LLP}}{v_{LLP}} + \frac{l_b}{c} - t_{expected}$$

Clustered
calorimeter hits

LLP

Trackless

LLP jet

2.95 m

HICAL

1.77 m

ECAL

1.29 m

Tracker

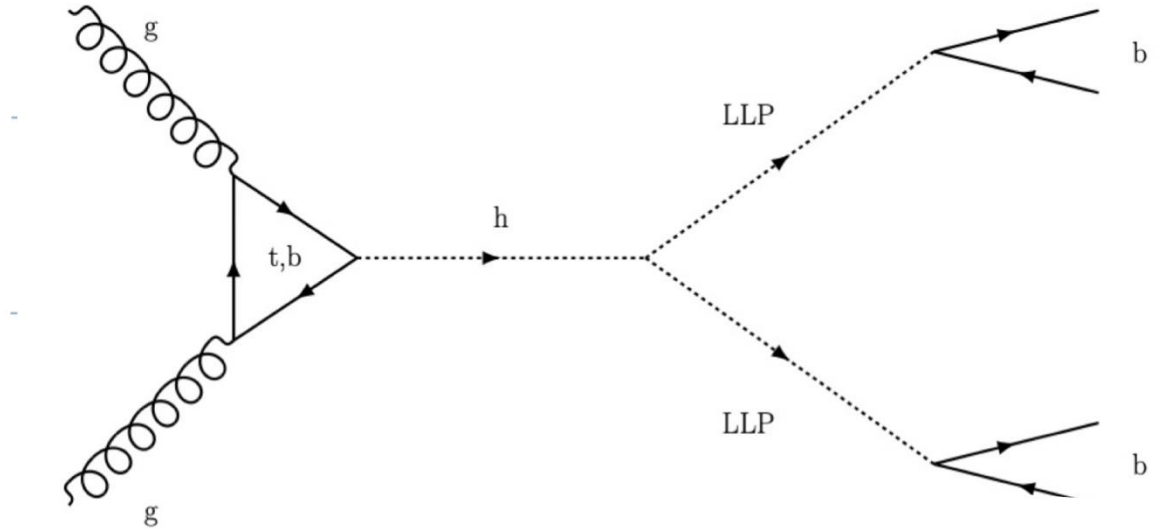
Search Strategy

- Searching for $H \rightarrow ss \rightarrow bbbb$
- Specifically within the HCAL: looking at HCAL decays

Online

Developed Level 1 Trigger to select LLP-matching events:

- Higher energy fraction in HCAL segmentation
- Time delay
- Jet Energy threshold 40GeV
- Selection of $\approx 100\text{M}$ events in 2023

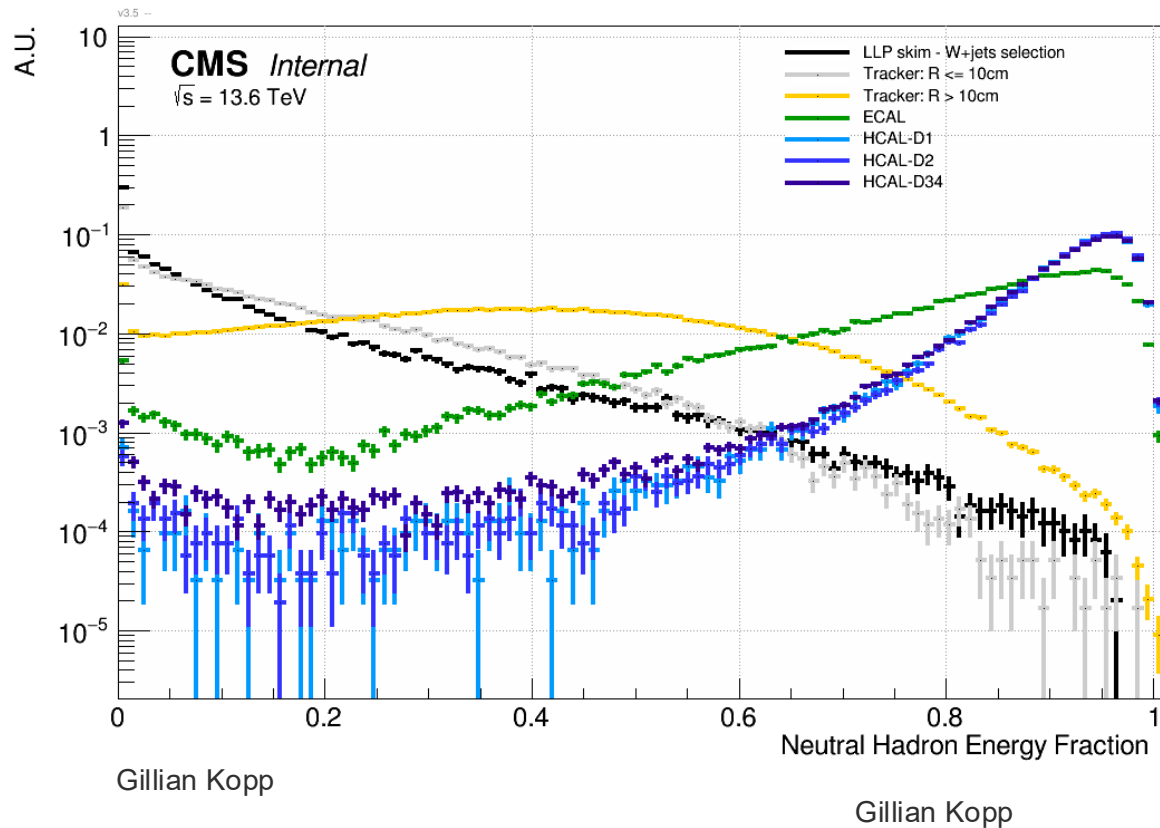


Offline Analysis

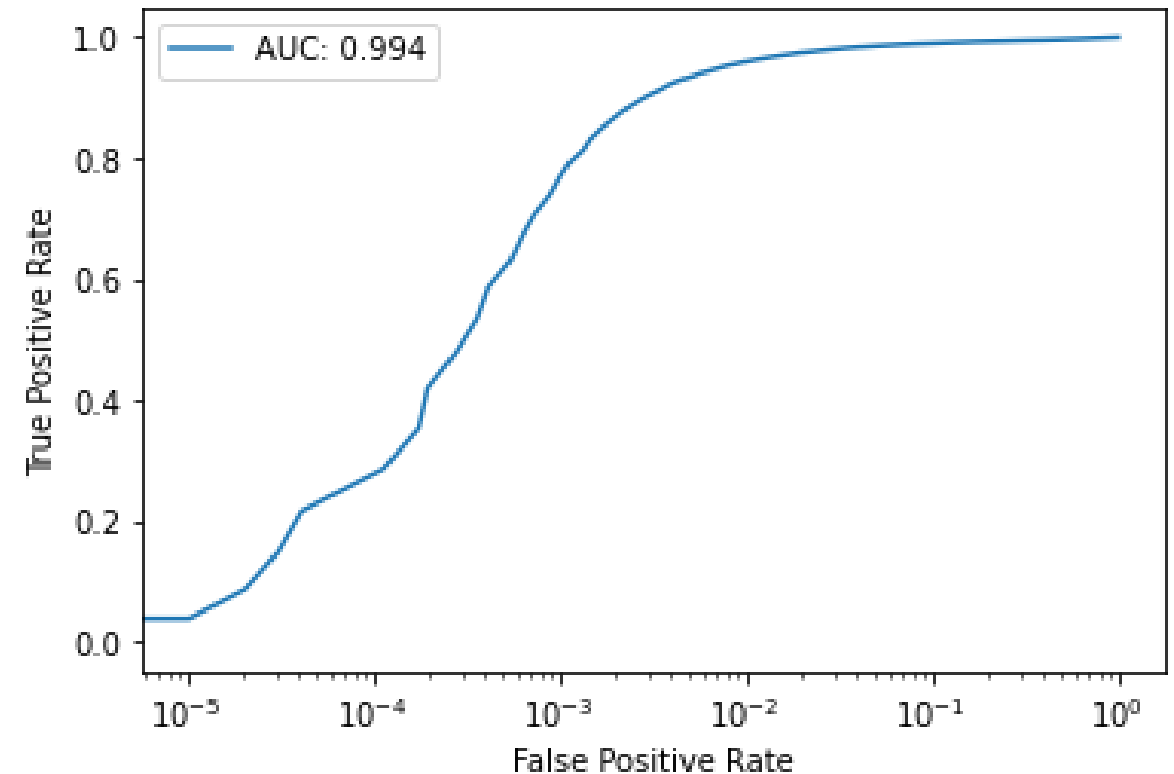
- Develop a powerful tagger/ classifier for selection of LLPs and rejection of background events

Sensitive Jet Object Variables

Jet: Neutral Hadron Energy Fraction



ROC Derived from perjet_NeutralHadEFrac Distribution



Data (for training)

Signal

- MC-generated LLP events, based on different combinations of Higgs Mass and LLP Mass; Each file contains roughly 100k entries (reduces to 30-70k after pre-selections)

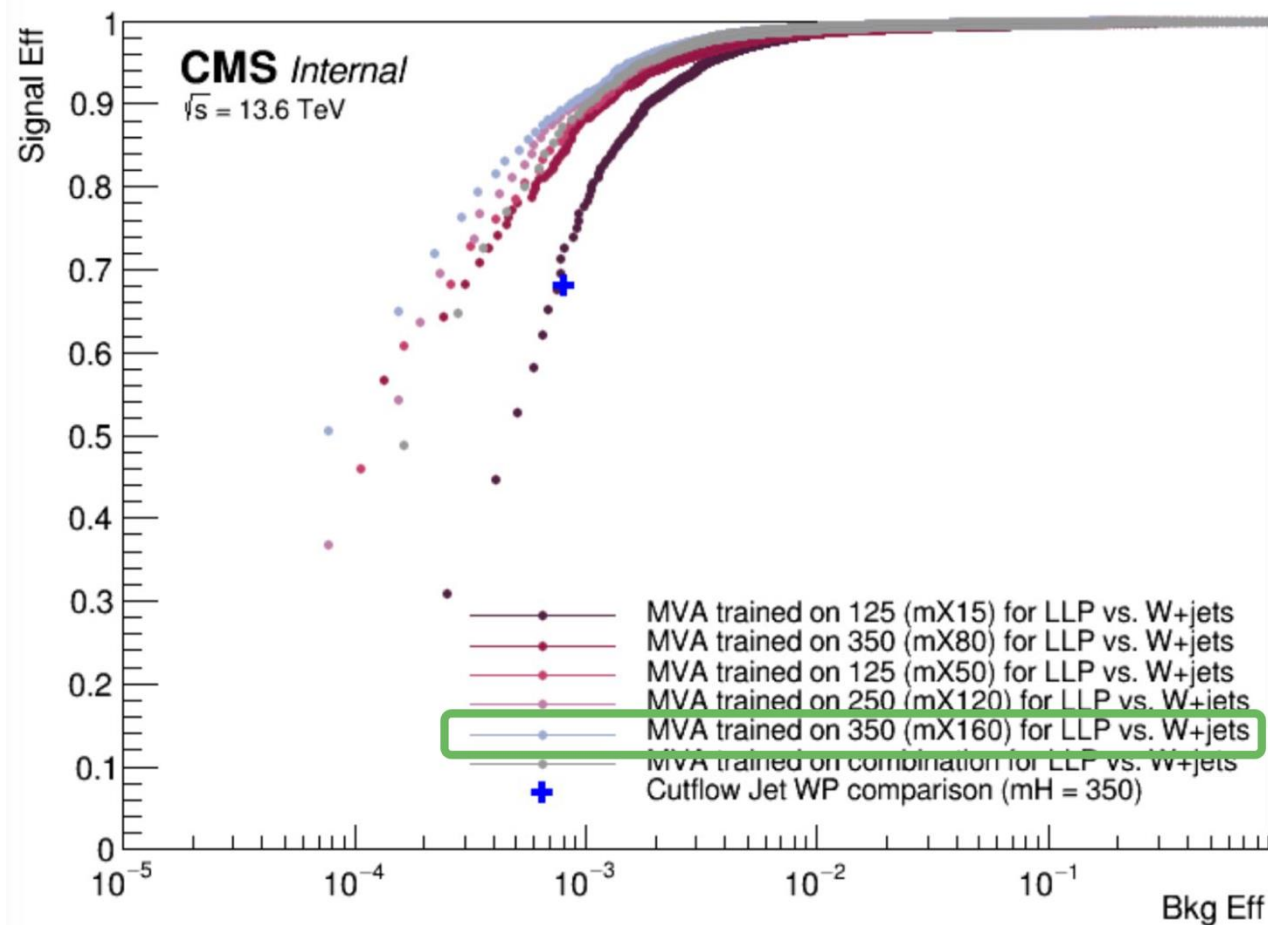
Background

- Run3 Skim for W^+ Jets $\approx$ 80M events
- Training only performed against W^+ jets (one source of background)

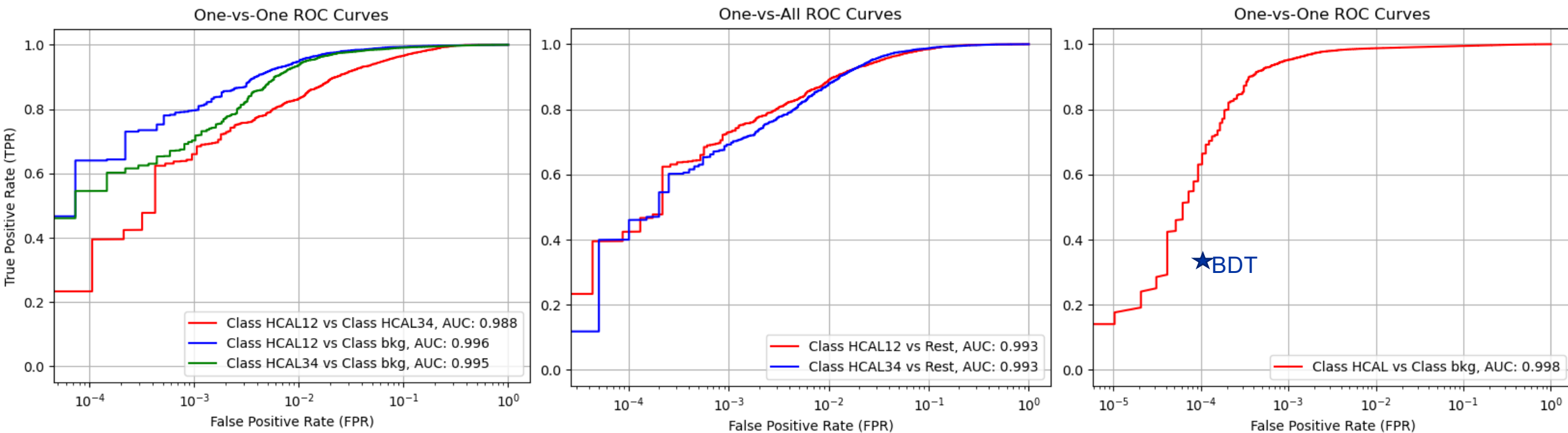
Higgs Mass [GeV]	LLP Mass [GeV]
125	15
125	15
125	15
125	50
125	50
250	120
350	80
350	160

Multi-Variate Analysis Framework: Gradient-Boosted Decision Tree

Evaluated on $M(H,X)=350,160$ GeV

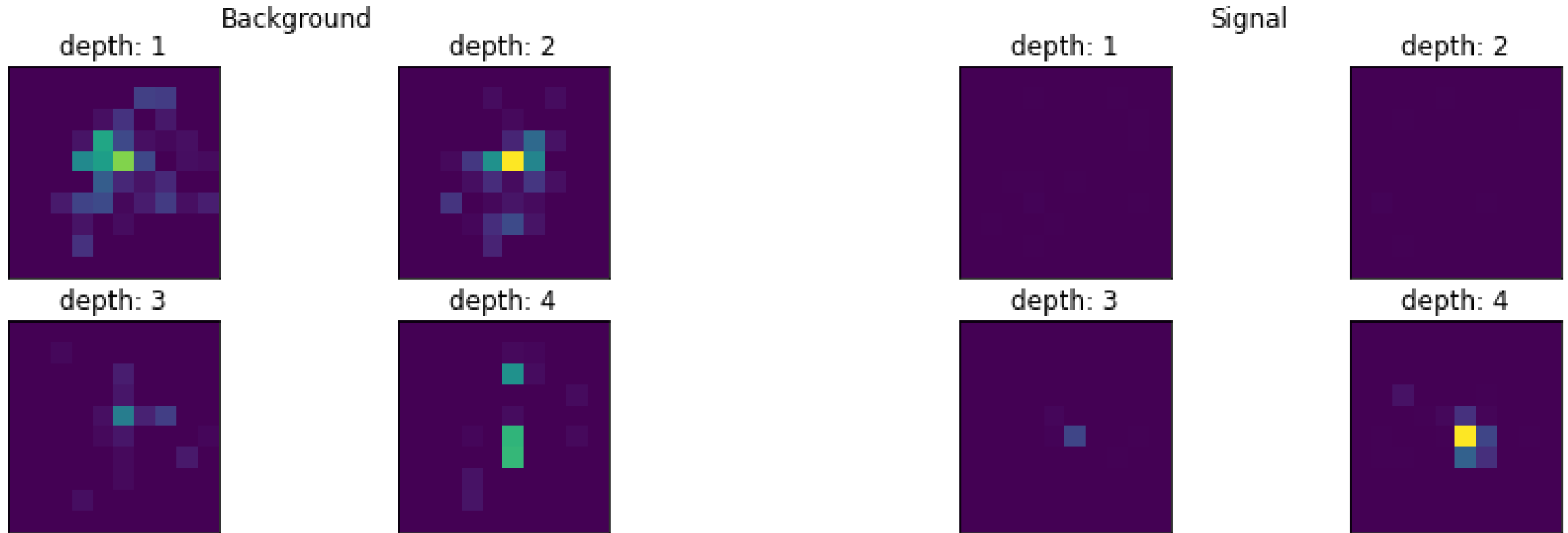


Dense NN (Multi-)Classifier: MH350-MS160



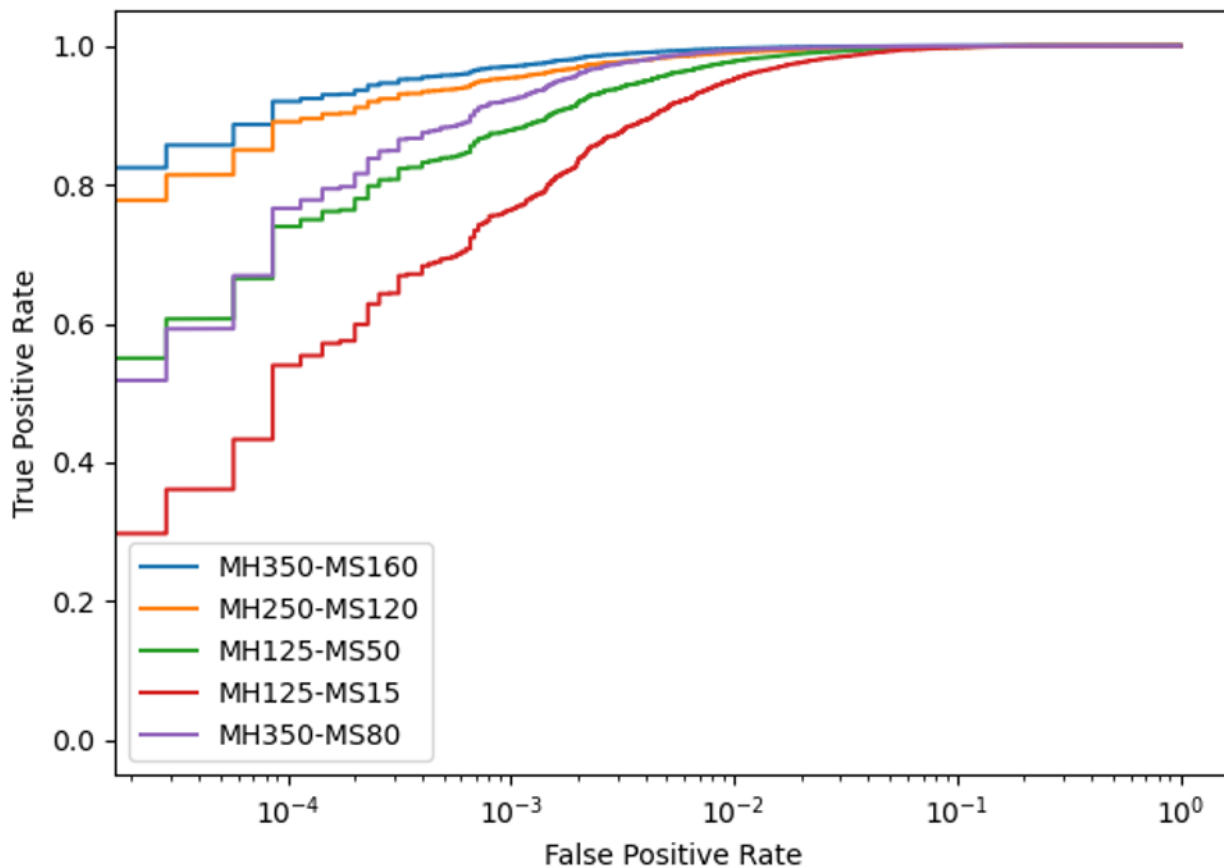
Training on 3D Jet Images

3D Image Sample from Signal and Background

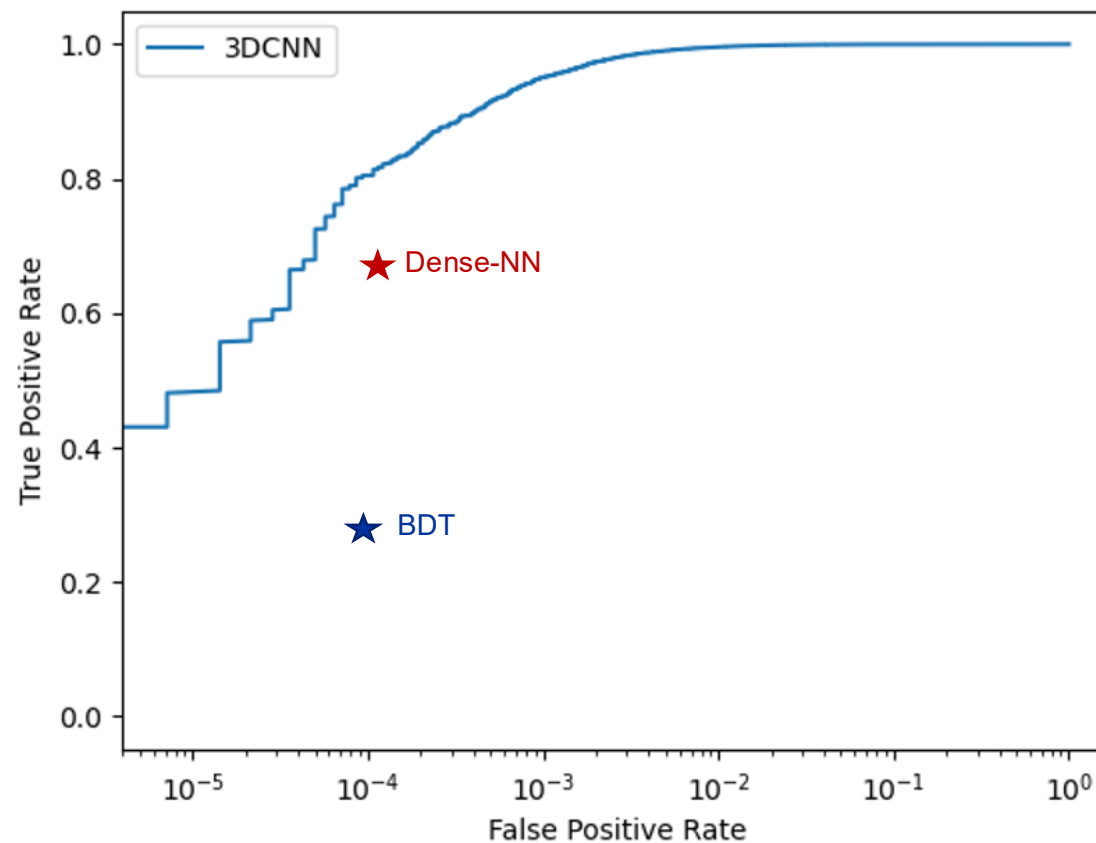


3DCNN Results

Trained on MH350-MS160 with 3DCNN

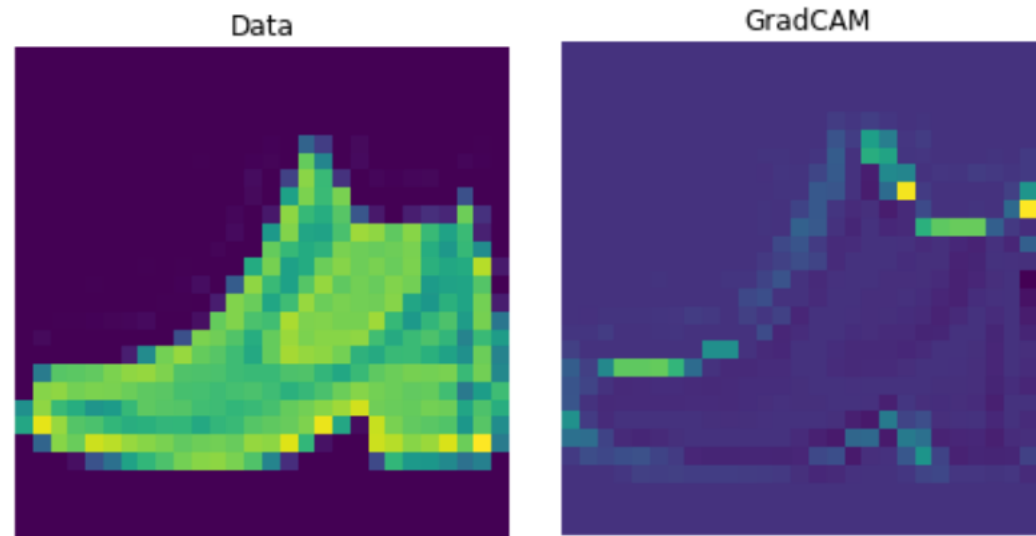


Evaluated on All Masses, Augmented Dataset



GradCAM: Gradient-Weighted Class Activation Mapping

- Pass image through last convolution layer (extracts highest level features)
- Take gradient of final output score with respect to each input pixel using backpropagation
- Apply ReLU and use gradients to weight importance of filters/ feature maps to Conv layer output when taking a weighted average to obtain a single output map



Applying GradCAM to Jet Images

Predicted and Labelled as: Background

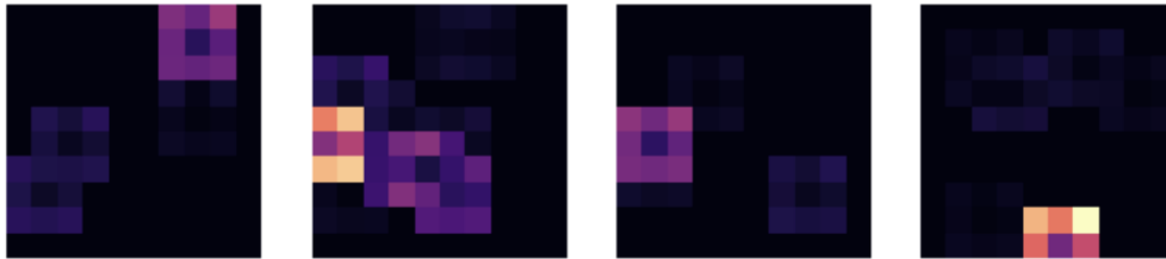


Depth1

Depth2

Depth3

Depth4



Signal GradCAM:

- Activation pattern mimics the shape of the jet cluster
- Model perceives nearby hot towers as one energy cluster

Background GradCAM:

- Model sees only individual, localized hot towers
- Even when towers are close-by, an overlap of the 3x3 kernel shape is seen
- Surroundings activated more than the center

Predicted and Labelled as Signal



Depth1

Depth2

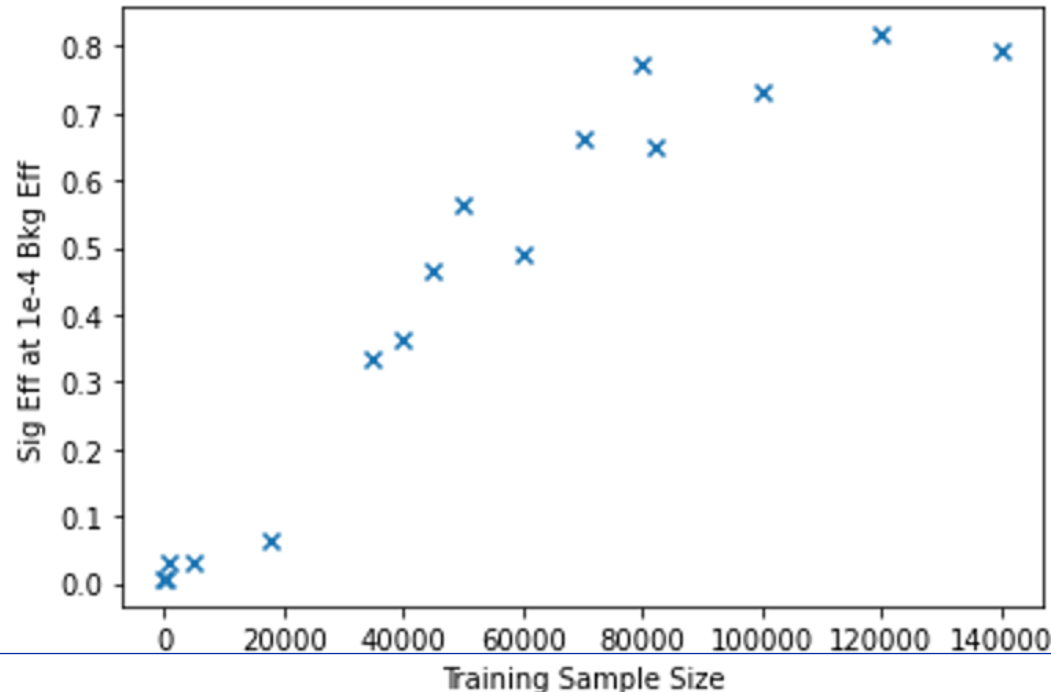
Depth3

Depth4



Conclusion and Future Steps

- Promising exploration of machine learning-based methods for tagging delayed jets and identifying long-lived particles.
- Each model exhibits some degree of interpretability.
- Most notable achievement so far is a 90% signal efficiency at just 0.01% background efficiency (3D Jet Image tagger).
- Possible exploration of more advanced classification methods



Plots of TPR evaluated at FPR = 1e-4 vs the amount of training data provided (evaluating samples are constant)

Thank You



**I am grateful to Professor Harvey Newman and Kiley Kennedy for their mentorship
and I thank the Samuel P. and Frances Krown SURF Fellowship**